Stream Introduction and FOSS license scanning:

The why, the how and the community approach

fossology

OSSelot
THE OPEN SOURCE CURATION DATABASE

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Working with FOSSology

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Agenda

Topic	Topic Name
1	What is FOSSology
2	Overall workflow
3	License identification
4	Processing copyrights
5	Reporting

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- This presentation concentrates on the interaction of users with a running FOSSology instance
- It is not a legal advice
- I am not a lawyer
- In case you have doubts please consult your lawyers

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What is FOSSology

- FOSSology is an open source license compliance software framework built to support a wide variety of use cases.
  - As a toolkit you can run license, copyright, export control and IP scans from the command line.
  - As a system, it is a server providing a database, web UI and a REST API for a customizable compliance workflow.
  - It provides a lot of different report file formats:
    - Text
    - CycloneDX
    - SPDX-2.3
    - SDPX-3.0
    - MS Word
    - OSSelot SPDX
  - Many features to reduce review tasks are implemented
    - Reuse of already made bulk actions
    - Import report
    - Reuse deactivated copyrights

fossology

https://github.com/FOSSology/FOSSology/
License: GPL-2.0-only

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What is FOSSology

- FOSSology provides for license identification several agents:
  - Nomos
  - Monk
  - Ojo
  - Scancode
- For copyrights you can use:
  - FOSSology agent
  - Scancode
- It provides also a keyword agent to search for specific keywords
  - E.g. Patrick McHardy or kaber@trash.net
- It provides also:
  - IP agent
  - ECC agent
- It provides a REST API. A ready to use python library to make use of the REST API is available at GitHub: https://github.com/fossology/fossology-python

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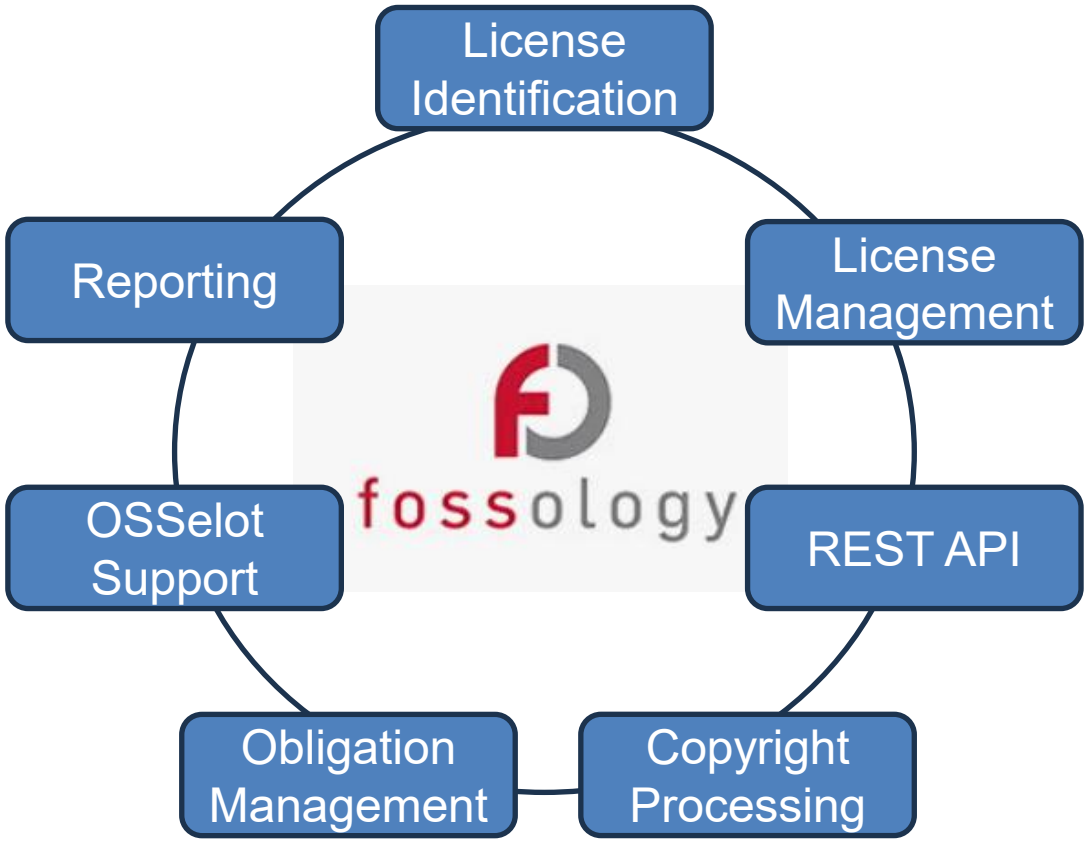
What is FOSSology

- FOSSology provides license management
  - Creating/ modifying new licenses
  - Creating license candidates
  - Assigning risk levels
  - Creating / modifying compatibility rules

- FOSSology provides obligation management
  - Creating/ modifying new obligations
  - Assigning obligations to licenses
  - Assigning obligations to license candidates

- FOSSology allows to store required acknowledgments in the database



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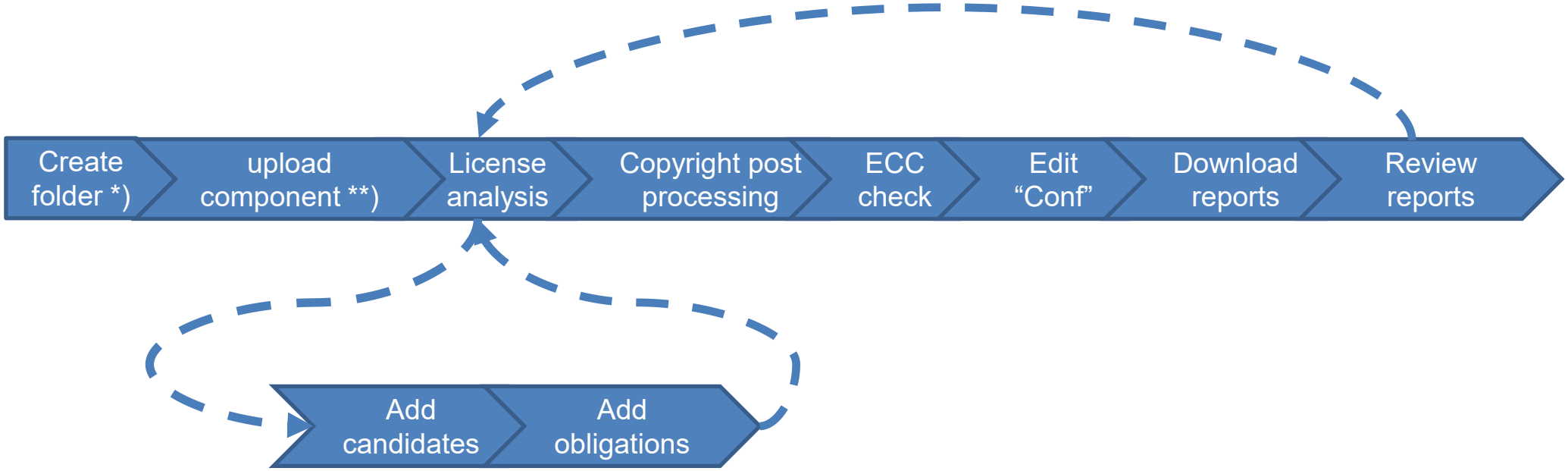
The Overall Workflow



*) Create the right folder if not yet existing
**) Upload the component to the right folder – reuse bulk phrases if possible

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Upload a Source Code Package

fossology

Open Source License Compliance by Open Source Software

1. Select the folder for storing the uploaded files:

   Software Repository

2. Select the file(s) to upload:

   Dateien auswählen Keine ausgewählt

3. Description(s)
   Select file(s) to enter description

4. ☐ Apply global decisions for current upload

5. ☐ Ignore SCM files (Git, SVN, TFS) and files with particular Mimetype

6. ☐ Ignore Configured Folders: tests/ × .github/ × examples/ ×

7. ○ Visible only for active group
   ◉ Visible for all groups
   ○ Make Public

8. Select optional analysis:
   - ☐ Compatibility License Analysis, scanning for licenses compatibility
   - ☐ Copyright/Email/URL/Author Analysis
   - ☐ ECC Analysis
   - ☐ IPRA Analysis
   - ☐ Keyword Analysis
   - ☐ MIME-type Analysis
   - ☐ Monk License Analysis
   - ☐ Nomos License Analysis
   - ☐ Ojo License Analysis
   - ☐ Package Analysis
   - ☐ REUSE.Software Analysis
   - ☐ SCANOSS Toolkit
   - ☐ Software Heritage Analysis

9. Automatic Concluded License Decider , based on
   - ☐ Scanners matches if all Nomos findings are within the Monk findings
   - ☐ Scanners matches if Ojo or REUSE.Software findings are no contradiction with other findi
   - ☐ Bulk phrases from reused packages
   - ☐ New scanner results, i.e., decisions were marked as work in progress if new scanner find
   - ☐ Auto conclude license finding if they are of type Permissive
   - ☐ False Positive Copyright Deactivation
   - ☐ False Positive Copyright Deactivation and clutter removal

10. (Optional) Reuse

    Set the reuse information

11. ScanCode Toolkit scan for
    - ☐ License
    - ☐ Copyright
    - ☐ Email
    - ☐ URL

After you press Upload, please be patient while your file is transferring.

Upload

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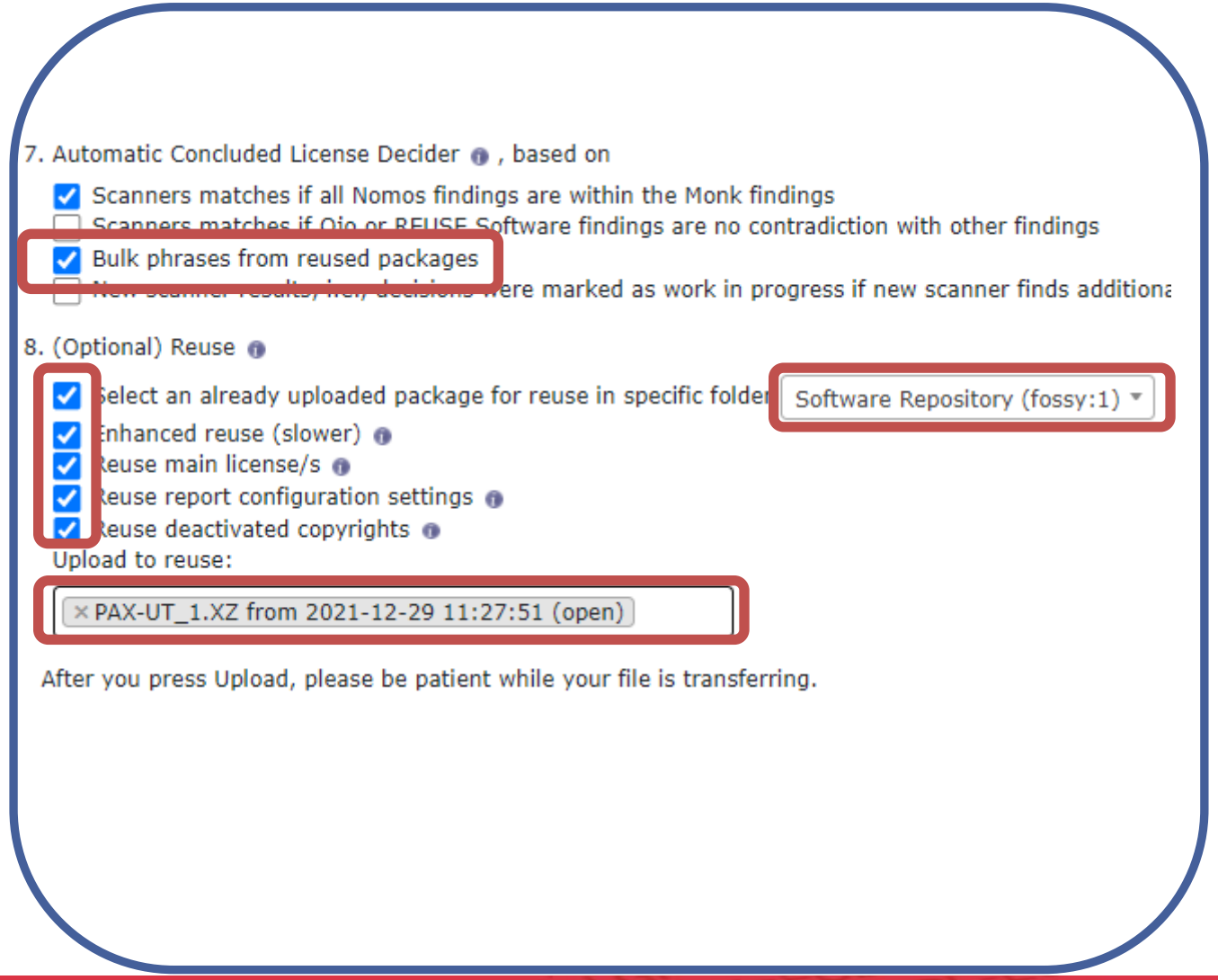
Reusing Bulk Phrases on newer Releases

- FOSSology stores all the applied bulk identification jobs.
- They can be reused and applied to new uploads. This ensures that already done identifications must not be repeated
- You can reuse not only the bulk identification but also:
  - The main license
  - The Conf section
  - And the copyright statements
- Note: Reusing bulk identification jobs put load on the server, but better load on the server than repeated work for humans



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Excluding folders from scanning

When uploading from file
Select „Ignore Configured Folders“
Deselect those from the list you still want to be included

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Excluding folders from scanning

Scan result with ignored folders

Scanner Count	Concluded Count	License Name
139	0	GPL-2.0-or-later
139	0	Apache-2.0
138	0	Dual-license
1	0	UnclassifiedLicense
1	0	See-file
1	0	See-doc.OTHER
1	0	See-URL
1	0	GPL-2.0-only
1	0	BSD-3-Clause

Showing 1 to 9 of 9 licenses

Scan result without ignored folders

Scanner Count	Concluded License Count	License Name
223	218	Apache-2.0
222	217	GPL-2.0-or-later
221	217	Dual-license
2	0	See-file
2	0	BSD-3-Clause
1	2	License-of-GNU-Licenses
1	0	UnclassifiedLicense
1	0	See-doc.OTHER
1	0	See-URL
1	0	LicenseRef-scancode-unknown-license-reference
1	0	LicenseRef-scancode-unknown
1	0	LicenseRef-scancode-free-unknown
1	0	LicenseRef-scancode-dco-1.1
1	0	GPL-2.0-or-later WITH Autoconf-exception
1	0	GPL-2.0-only

Showing 1 to 15 of 15 licenses

Page 1

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License Identification

- Identifying the correct license of a file is the most important task during package analysis.

- FOSSology provides many features to support this task and to allow for efficient work
  - Different license scanners are implemented:
    - Nomos
    - Monk
    - Ojo
    - Scancode
- Text highlighting for fast identification of license text modifications

- Usually, the scanner findings and the concluded licenses are different after finishing the license identification

- Matched license overview for immediate response to the development, without the need of a full package analysis

Scanner Count	Concluded License Count	License Name
913	0	GPL-3.0-or-later
46	0	GPL-2.0-or-later
46	0	FSFULLR
41	0	LicenseRef-scancode-free-unknown
33	0	GPL-1.0-or-later
32	0	GPL-3.0-only
30	0	LicenseRef-scancode-warranty-disclaimer
28	0	LicenseRef-scancode-fsf-notice
10	0	GPL-2.0-only
9	0	NOT-public-domain
8	0	Gettext-Permission-Notice
8	0	GFDL-1.3-or-later
8	0	GFDL-1.1-or-later
6	0	LicenseRef-scancode-fsf-unlimited-no-warranty
6	0	GFDL-1.3-only
6	0	GFDL-1.3
5	0	LicenseRef-scancode-public-domain
4	0	Public-domain-ref
4	0	Public-domain
4	0	LicenseRef-scancode-other-permissive
4	0	Latex2e
4	0	BSD-3-Clause
3	0	UnclassifiedLicense
3	0	GPL
3	0	GNU-Manpages

Showing 1 to 25 of 64 licenses

Page 1 of 3

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License Identification

- FOSSology provides different ways to identify the license of a file:
  - Manual inspection file by file
  - Batch identification via import report
  - License assignment on folder level
  - Bulk scans

- You can provide:
  - Individual license texts
  - Acknowledgements
  - Comments

Action	License	Source	License Text	Acknowledgement	Comment
	Permission-Notice	Bulk: #9194882	Permission to u...	Click to add	Click to add
	HPND	scancode: #1 (100 %)	-	-	-
	ISC	nomos: #1	-	-	-
	BSD-4-Clause-UC	nomos: #1 scancode: #1 (100 %) monk: #1 (97 %) Bulk: #9194885	Redistribution ...	Click to add	The BSD-4-Claus...

Showing 1 to 4 of 4 entries

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License Identification

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Scanner Count	Concluded License Count	License Name
913	0	GPL-3.0-or-later
46	0	GPL-2.0-or-later
46	0	FSFULLR
41	0	LicenseRef-scancode-free-unknown
33	0	GPL-1.0-or-later
32	0	GPL-3.0-only
30	0	LicenseRef-scancode-warranty-disclaimer
28	0	LicenseRef-scancode-fsf-notice
10	0	GPL-2.0-only
9	0	NOT-public-domain
8	0	Gettext-Permission-Notice
8	0	GFDL-1.3-or-later
8	0	GFDL-1.1-or-later
6	0	LicenseRef-scancode-fsf-unlimited-no-warranty
6	0	GFDL-1.3-only
6	0	GFDL-1.3
5	0	LicenseRef-scancode-public-domain
4	0	Public-domain-ref
4	0	Public-domain
4	0	LicenseRef-scancode-other-permissive
4	0	Latex2e
4	0	BSD-3-Clause
3	0	UnclassifiedLicense
3	0	GPL
3	0	GNU-Manpages

Showing 1 to 25 of 64 licenses

Page 1 of 3

Scanner Count	Concluded License Count	License Name
913	874	GPL-3.0-or-later
46	41	FSFULLR
46	29	GPL-2.0-or-later
41	0	LicenseRef-scancode-free-unknown
33	0	GPL-1.0-or-later
32	0	GPL-3.0-only
30	0	LicenseRef-scancode-warranty-disclaimer
28	0	LicenseRef-scancode-fsf-notice
10	0	GPL-2.0-only
9	0	NOT-public-domain
8	9	Gettext-Permission-Notice
8	0	GFDL-1.3-or-later
8	0	GFDL-1.1-or-later
6	0	LicenseRef-scancode-fsf-unlimited-no-warranty
6	0	GFDL-1.3-only
6	0	GFDL-1.3
5	0	LicenseRef-scancode-public-domain
4	5	Public-domain
4	1	Latex2e
4	0	Public-domain-ref
4	0	LicenseRef-scancode-other-permissive
4	0	BSD-3-Clause
3	3	GNU-Manpages
3	3	FSFAP
3	0	UnclassifiedLicense

Showing 1 to 25 of 72 licenses

Page 1 of 3

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License Identification - Compatibility

FOSSology provides license compatibility rules and checks

Per default FOSSology comes with an empty rule set

You can define own compatibility rules and you can import the OSADL compatibility list: https://www.osadl.org/fileadmin/checklists/matrixseq.json

hilbert.cpp	**GPL-2.0-or-later** [N][O][Sc: 100%], **Apache-2.0** [N][O][Sc: 100%], **BSD-4-Clause** [N][O][Sc: 100%], URT-RLE [Sc: 73%], GPL-1.0-or-later [Sc: 100%], GPL-2.0-only [Sc: 100%]
init.cfg	GPL-3.0-or-later [N][Sc: 100%]
j2ee_web_services_1_1.xsd	Sun-IP [N], LicenseRef-scancode-proprietary-license [Sc: 100%]
Makefile.txt	BSD-style [N], BSD-4-Clause [M: 80%], Caldera-no-preamble [Sc: 100%]
nir_vla.h	MIT [N][M: 89%][Sc: 99%]
node.c	BSD-3-Clause [N][Sc: 100%], BSD-3-Clause_ATMEL [M: 89%]

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License Identification Real vs. License Text in the Database

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*
```

Action	License	Source	License Text	Acknowledgem
✖ ☆	HPND-sell-variant	nomos: #1 scancode: #1 (99 %) monk: #1 (90 %) Bulk: #10428669	Permission to u...	Click to add

Showing 1 to 1 of 1 entries

Historical Permission Notice and Disclaimer - sell variant

Reference URL:

https://git.kernel.org/pub/scm/linux/kernel/git/torvalds/linux.git/tree/net/sunrpc/auth_gss/gss_generic_token.c?h=v4.19

Risk level:

0

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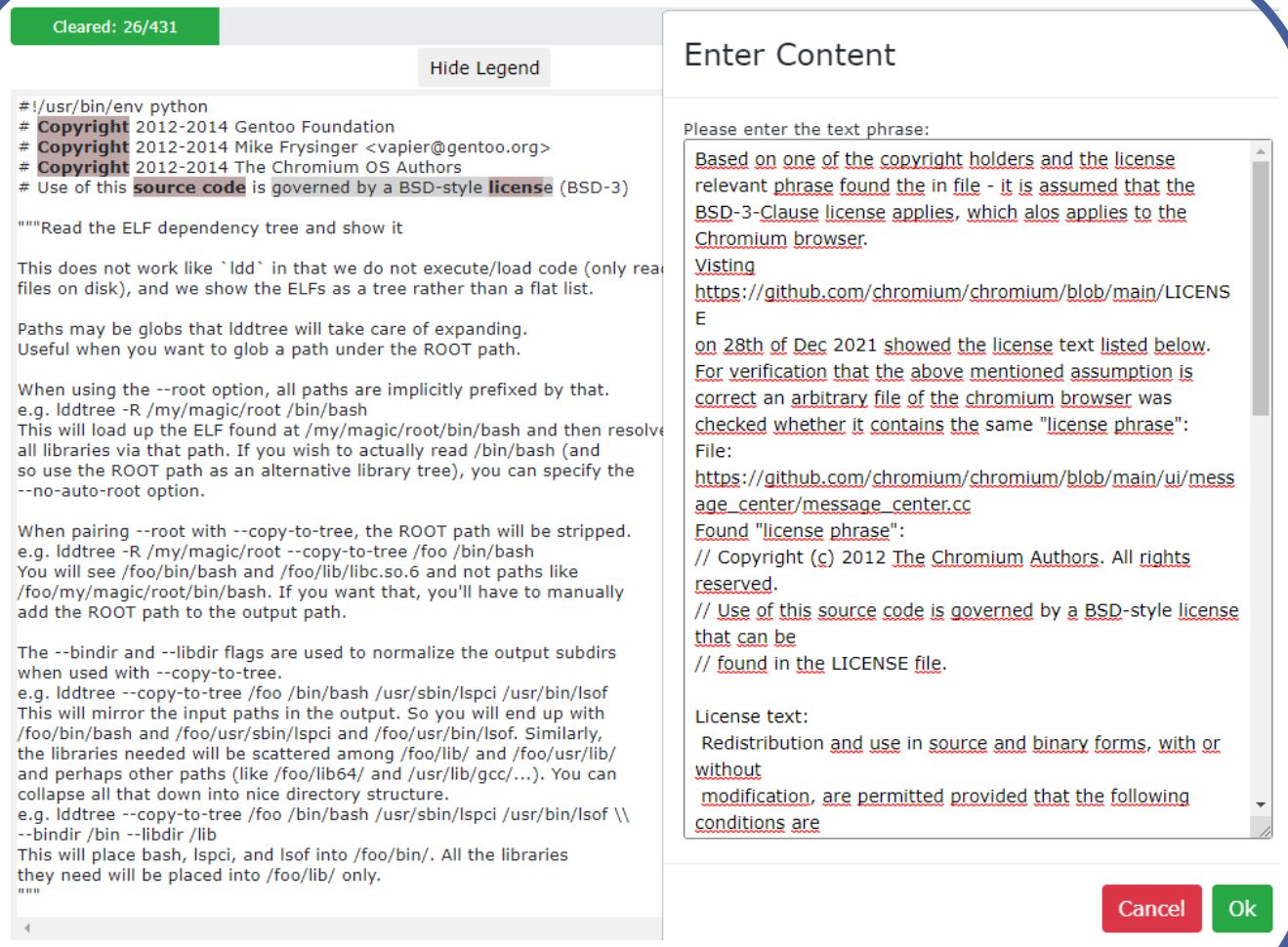
- Sometimes it is necessary to add information about why a certain license conclusion was made. Use the comment field for this task and provide at least the following information:
  - What was done (e.g. surfing to the site https://github.com/chromium/chromium/blob/main/LICENSE
  - What was displayed (e.g. copy the license text)
  - Date when the site was visited

  The comment text fields are generated in the “Unified Report” and if configured in the SPDX tag value files. Thus, you have an audit trail why some conclusions were done



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License Identification Providing Individual License Texts

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Home Search Browse Upload Jobs Organize Admin Help

Change concluded License

Version: [4.4.0.51], Branch: [osadl-osselot], Commit: [#1e52cb] 2024/05/24 16:02 +02:00 built @ 2024/06/07 10:12 +02:00

Folder: Software Repository/ ofendt/ libxkbcommon/ libxkbcommon-xkbcommon-1.7.0.tar.gz/libxkbcommon-xkbcommon-1.7.0.tar/libxkbcommon-xkbcommon-1.7.0

Show Legend

```
# Derived from: https://github.com/ilanschnell/perfect-hash
# Commit: 6b7dd80a525dbd4349ea2c69f04a9c96f3c2fd54

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```

Clean Text | Cancel | Ok

enses | Copyright/Email/Url/Author | ECC | IPRA |

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licenses and no clearing result

this file

			License Text
	BSD-3-Clause-HP	monk: #1 (92 %)	-
	BSD-3-Clause	nomos: #1 scancode: #1 (98 %) Bulk: #10427275	Redistribution ...

Showing 1 to 2 of 2 entries

License Identification Searching for the Correct License Text

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```
tial keypad_keys
)_symbols "ops" {
  key <KPDV> { [ NoSymbol,  slash,     NoSymbol,  XF86_Ungrab     ] }
  key <KPMU> { [ NoSymbol,  asterisk,  NoSymbol,  XF86_ClearGrab  ] }
  key <KPSU> { [ NoSymbol,  minus,     NoSymbol,  XF86_Prev_VMode ] }
  key <KPAD> { [ NoSymbol,  plus,      NoSymbol,  XF86_Next_VMode ] }
  key <KPEQ> { [ NoSymbol,  equal      ] };

  Hexadecimal Numpad, by Roland Kaufmann <rlndkfmn at gmail dot com>
  License: BSD (also covers variant with ATM digit order)
  Third level gives the letters used in hexadecimal numbers, or
  columns in small spreadsheets. As a bonus, having 'e' available at
  the center of the keyboard makes it easy to enter floating point
  numbers in scientific notation.
  Equal is added to start formulas (most numpads on PCs don't have
  their own equal key), comma as a list separator (as most
  programming languages wants period as a decimal separator) and
  colon to enter times and ranges. Someone also may want to represent
  multiplication and division in prose using x and colon.
  Two first levels are specified as NoSymbol to not override any
  previous specification. Combine this with another keypad specifier,
  e.g. "legacy".
partial keypad_keys
)_symbols "hex" {
  key.type[Group1]="FOUR_LEVEL_MIXED_KEYPAD" ;
```

Clearing decision type

- ○ Identified
- ○ Irrelevant
- ○ Do not use

Clearing decision s

☐ Apply decision to all future o file

Action	License	Source	License Text
✖ ☆	BSD	nomos: #1	Click to add
✖ ☆	BSD-3-Clause	scancode: #1 (99 %)	Click to add

Showing 1 to 2 of 2 entries

User Decision ... | Bulk Recognition ... | Clearing History ...

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License Identification Searching for the Correct License Text

Sometimes the identification of the correct license of a file involves time and effort and cannot be automated.

Searching in the internet leads to:
https://www.kaufmann.no/roland/dvorak/

Check whether the information on the site relates to the part of the file in the package to analyze

The license which is finally displayed is a 5 clause license which is a permissive license

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Finally, the correct license can be concluded with a comment documenting all the executed steps

```
};

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// License: BSD (also covers variant with ATM digit order)
// Third level gives the letters used in hexadecimal numbers, or
// columns in small spreadsheets. As a bonus, having 'e' available at
// the center of the keyboard makes it easy to enter floating point
```

Text on the webpage

tever desires, but you can not call the re
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```
key <KPAD> { [ NoSymbol, plus, NoSymbol, XF86_Next_VMode ] };
key <KPEQ> { [ NoSymbol, equal ] };
};

// Hexadecimal Numpad, by Roland Kaufmann <rlndkfmn at gmail dot com>
// License: BSD (also covers variant with ATM digit order)
// Third-level gives the letters used in hexadecimal numbers, or
```

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- Identified
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In such cases you start searching the internet: Keywords: Hilbert Doug Moore

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The Readme contains:
####License From the code:

```
/* See LICENSE below for information on rights to use, modify and distribute
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 hilbert.c - Computes Hilbert space-filling curve coordinates, without
 recursion, from integer index, and vice versa, and other Hilbert-related
 calculations. Also known as Pi-order or Peano scan.

 Author Doug Moore
 Dept. of Computational and Applied Math
 Rice University
 httpwww.caam.rice.edu~dougm
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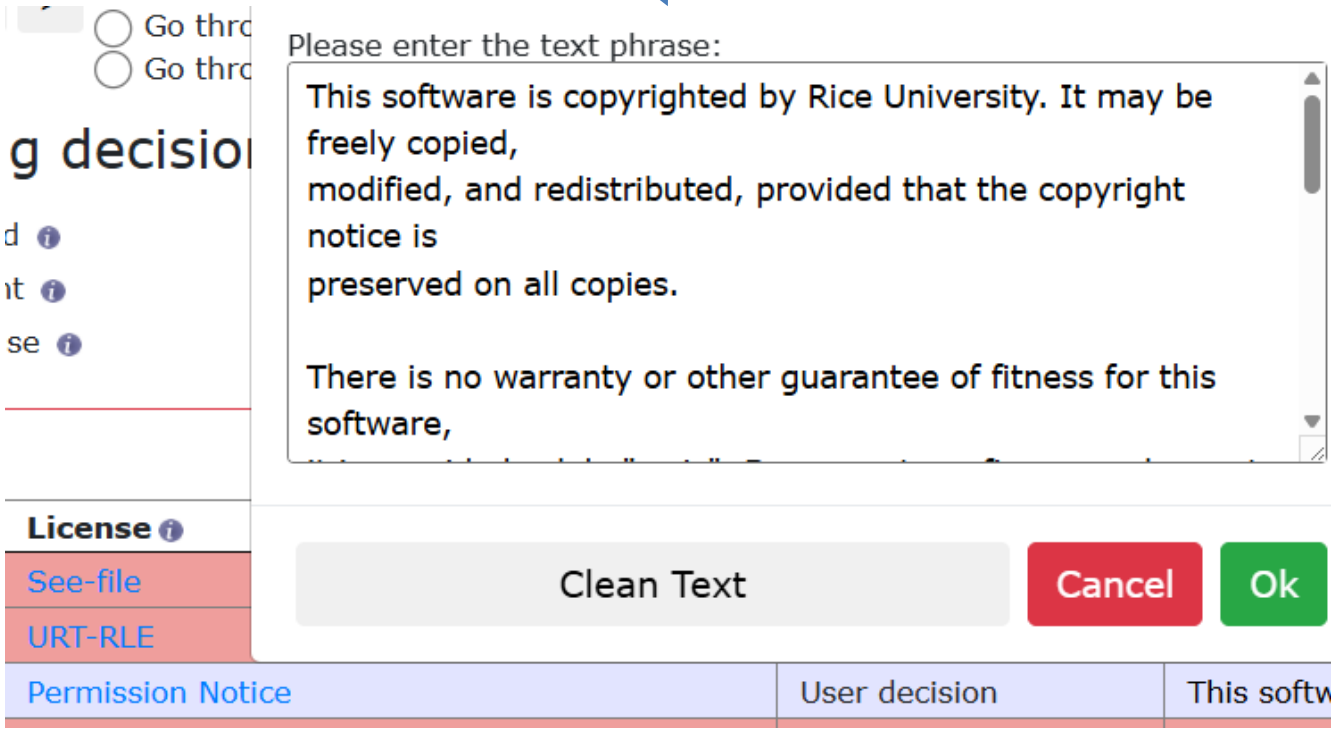

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# This file is a normative contributory data file in the
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#
# This file defines the Joining_Type and Joining_Group property
# values for Arabic, Syriac, N'Ko, Mandaic, and Manichaean positional
# shaping, repeating in machine readable form the information
# exemplified in Tables 9-3, 9-8, 9-9, 9-10, 9-14, 9-15, 9-16, 9-19,
# 9-20, 10-4, 10-5, 10-6, 10-7, and 19-5 of The Unicode Standard core
# specification. This file also defines Joining_Type values for
# Mongolian, Phags-pa, Psalter Pahlavi, Sogdian, Old Uyghur, Chora
# and Adlam positional shaping,
# and Joining_Type and Joining_Group values for Hanifi Rohingya pc
# which are not listed in tables in the standard.
#
# See Sections 9.2, 9.3, 9.5, 10.5, 10.6, 13.5, 14.4, 14.10, 14.11
# of The Unicode Standard core specification for more information.
#
# Each line contains four fields, separated by a semicolon.
#
```

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license text:
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#
# This file is a normative contributory data file in the
# Unicode Character Database.
#
# This file defines the Joining_Type and Joining_Group property
# values for Arabic, Syriac, N'Ko, Mandaic, and Manichaean positional
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# exemplified in Tables 9-3, 9-8, 9-9, 9-10, 9-14, 9-15, 9-16, 9-19,
# 9-20, 10-4, 10-5, 10-6, 10-7, and 19-5 of The Unicode Standard core
# specification. This file also defines Joining_Type values for
# Mongolian, Phags-pa, Psalter Pahlavi, Sogdian, Old Uyghur, Chora
# and Adlam positional shaping,
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#
# See Sections 9.2, 9.3, 9.5, 10.5, 10.6, 13.5, 14.4, 14.10, 14.11
# of The Unicode Standard core specification for more information.
#
# Each line contains four fields, separated by a semicolon.
#
```

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Agenda

Topic	Topic Name
1	What is FOSSology
2	Overall workflow
3	License identification
4	Processing copyrights
5	Reporting

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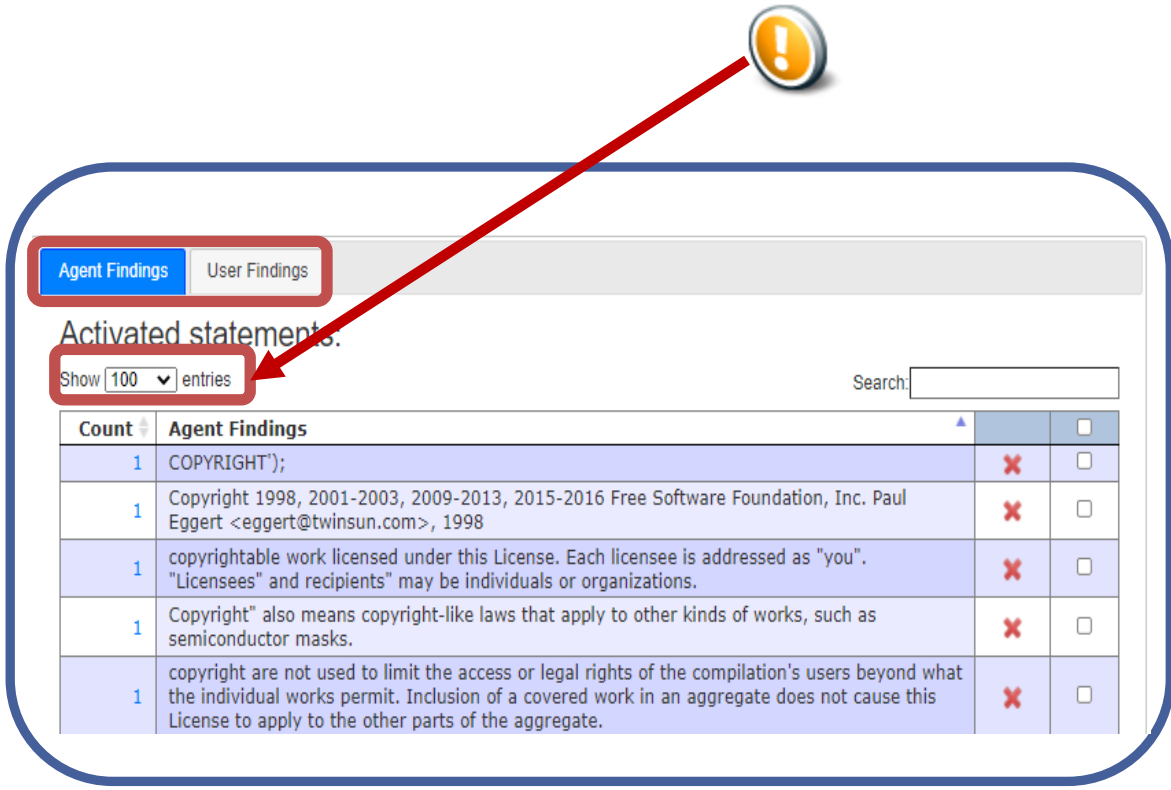
Copyright Statements

- FOSSology extracts copyright statements found in the source code files
- To achieve this, it uses regular expressions
- Usually, the found copyright statements need to be post processed and clutter need to be removed
- For post processing FOSSology provides two views:
  - The file view
  - The folder/upload view



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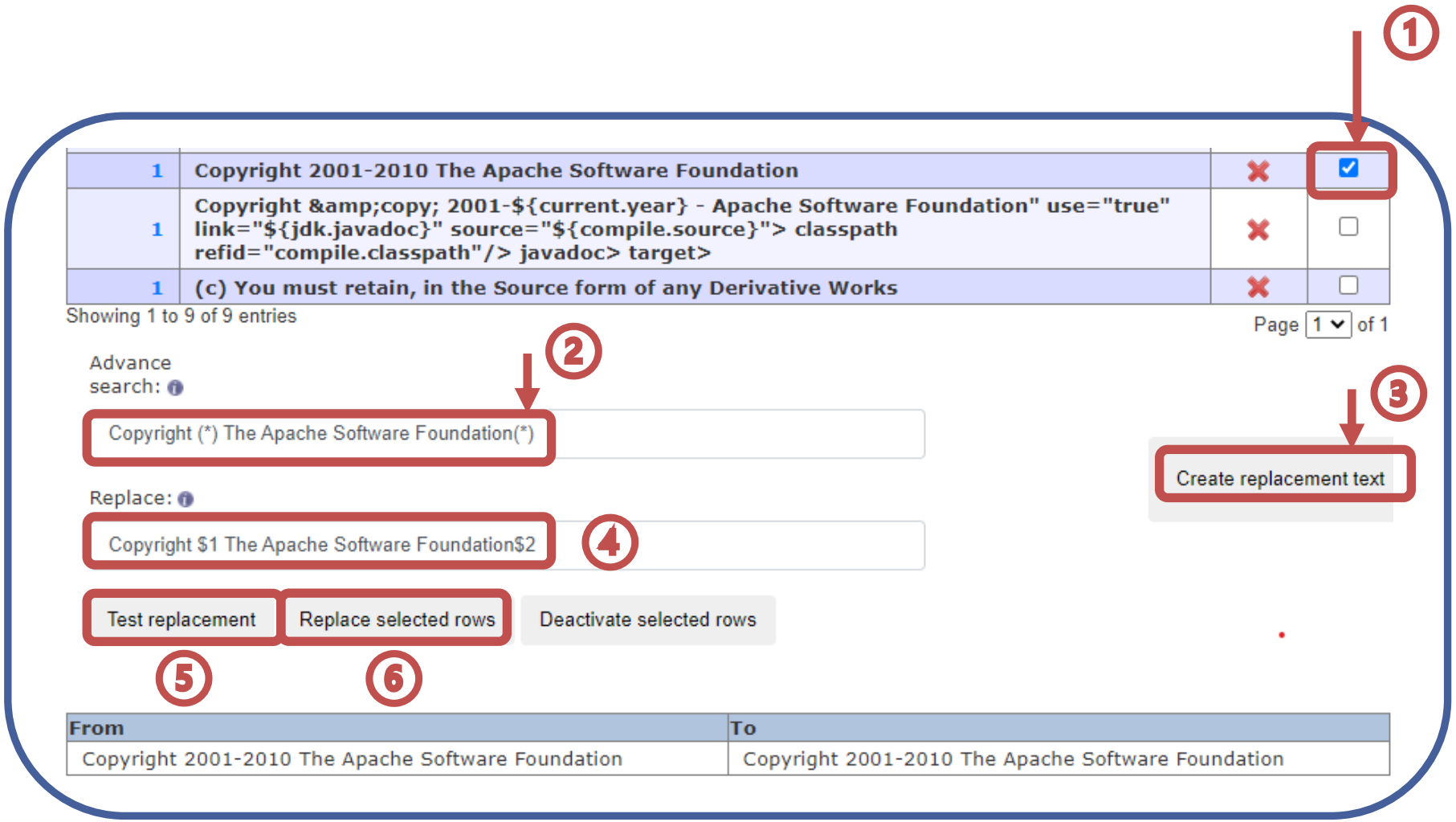

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Agenda

Topic	Topic Name
1	What is FOSSology
2	Overall workflow
3	License identification
4	Processing copyrights
5	Reporting

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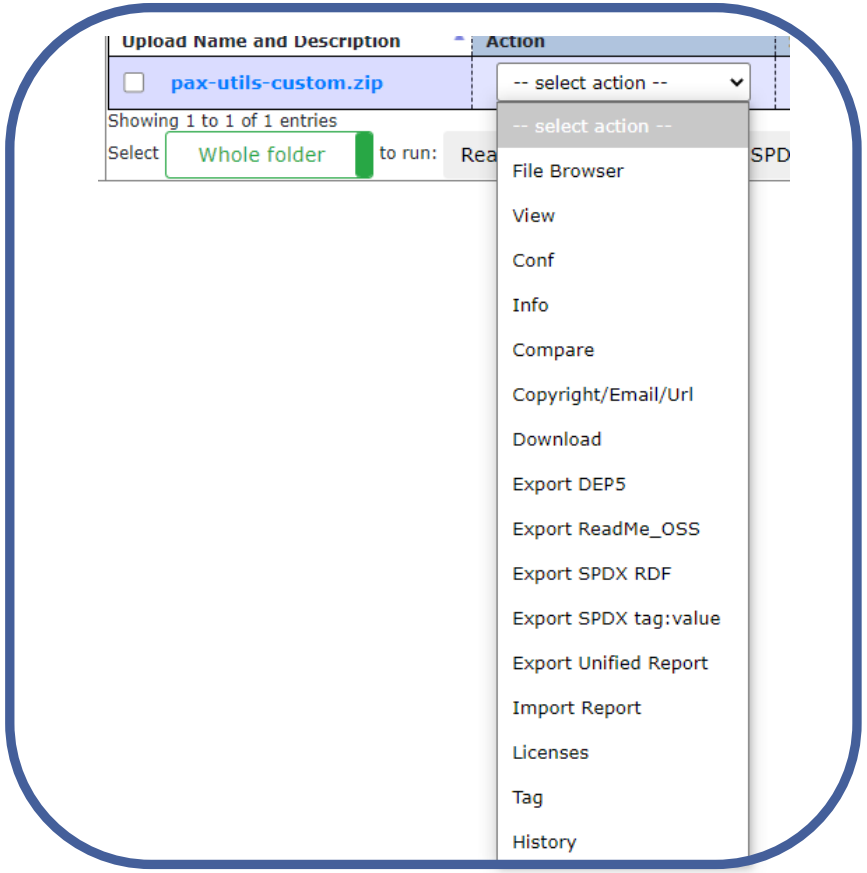
Reporting

Fossology provides many reporting features:

- Component level:
  - DEP5 files
  - ReadMe_OSS
  - SPDX V2
  - SPDX V3
  - CycloneDX
  - Unified Report
  - License List
  - Copyright List

- Folder level:
  - ReadMe_OSS
  - SPDX V2
  - SPDX V3
  - CycloneDX



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SPDXID: SPDXRef-item12773803
FileChecksum: SHA1: 0f773f388a00293b0
FileChecksum: SHA256: 7f8a09d0da266e5
FileChecksum: MD5: 84c48c6f1254e2d8c5
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```
FileName: fmt-10.1.1.tar.gz/fmt-10.1.
SPDXID: SPDXRef-item12773803
FileChecksum: SHA1: 0f773f388a00293b0
FileChecksum: SHA256: 7f8a09d0da266e5
FileChecksum: MD5: 84c48c6f1254e2d8c5
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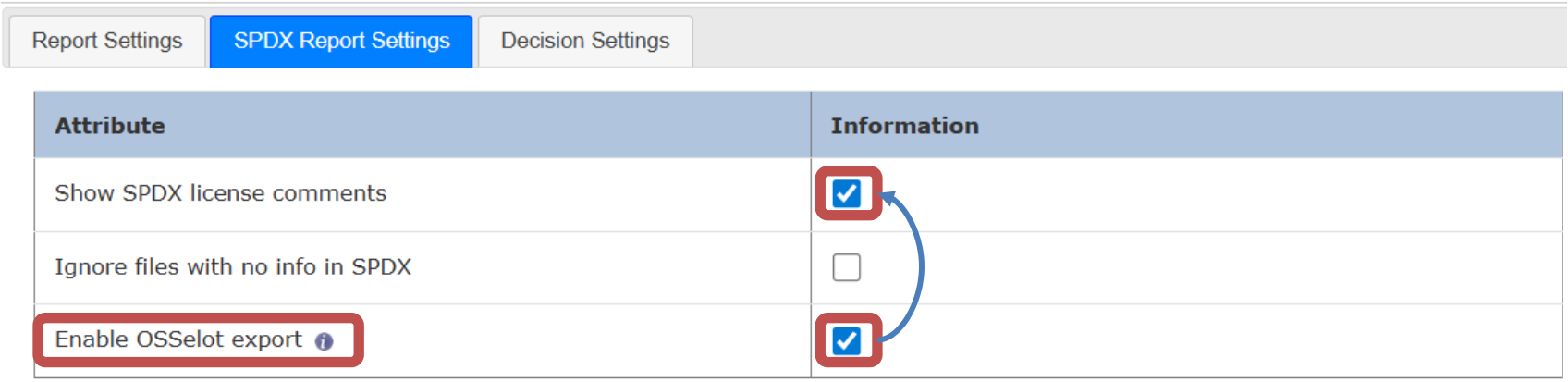

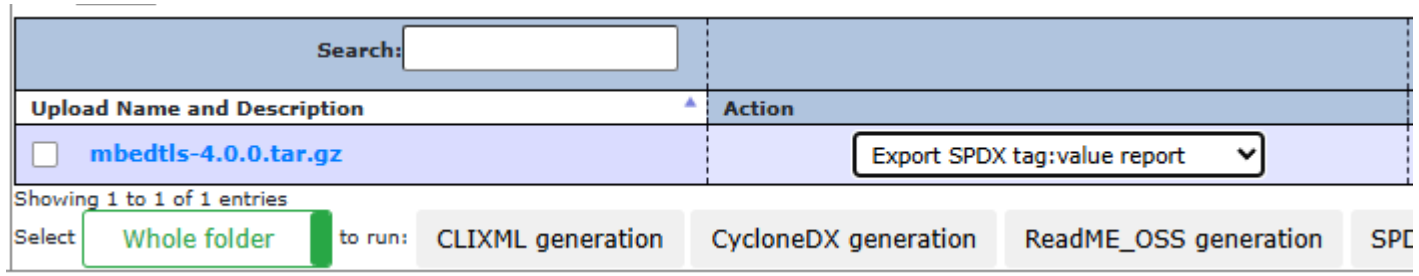

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Overall Take away

- FOSSology is a powerful tool and framework
- It provides many other useful features
- It is Open Source licensed under GPL-2.0
- As for any other powerful tool, a bit of a training is recommended to work with it in the most efficient way
- Installing FOSSology from the sources natively on a Linux machine is not difficult
- Proper license analysis cannot not be fully automated, at least not today

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